التصميم بمساعدة الحاسوب ح

418

Computer Graphics (Computer Aided Design)

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Lecture #1

Aims: The aims of this course are:

- **To provide the students with basic background of the Computer Graphics.**
- **To introduce the students with the knowledge of graphics tools.**
- **To provide the students with knowledge of how graphically 2D & 3D objects using graphics tools (OpenGL primitive tools).**
- **To introduce the students with concepts and implementations of the modeling transformations (2D & 3D).**
- **To provide the students with simple and non-simple mathematical models (tools) for objects modeling.**
- **Enabling the students to build a computer graphics applications based on OpenGL and the mathematical modeling graphical tools to produce and manipulate a computer graphics (images).**

Course contents:

- **History and introduction to computer graphics (CGs):**

  History of Computer Graphics, Applications of Computer Graphics, Introduction to Image and Objects, Image Representation, Scan-conversion algorithms of drawing computer graphics primitives (Lines).

- **Introducing the graphics software tool (OpenGL) under VC++:**

  Introduction to OpenGL, OpenGL Geometric Primitives, OpenGL 3D Models, OpenGL and Events Handling.

Course contents:

- **Model Transformation:**
  a. **Two Dimensional Transformations: Introduction to transformations, Transformation Matrix, Types of Transformations in Two-Dimensional Graphics: transilation, Scaling, Reflection, Homogeneous Coordinates, 2D Transformations using Homogeneous Coordinates**
  b. **Three-dimensional transformations: Objects in Homogeneous Coordinates, Three- Dimensional Transformations: Scaling, Translation, Rotation.**

- **Mathematical Modeling and OpenGL:**
  a. **Curves Design (2D Graphics): Curve Continuity, Parametric Curve, Bezier Curves, B-Spline Curves.**
  b. **Surfaces Design (3D Graphics): Bezier Surfaces, B-Spline Surfaces.**

References:

- **Computer Graphics with OpenGL, 3rd ed. (2004) by Doland Hearn and M. Pualine Baker.**
- **Computer Graphics using OpenGL 2nd ed. (2001) by F. S. Hill, Jr**
- **An introduction to NURBS (2001) by David F. Rogers.**
- **Open source - Internet (www.libgen.org)**

Assessment:

According to the college regulations

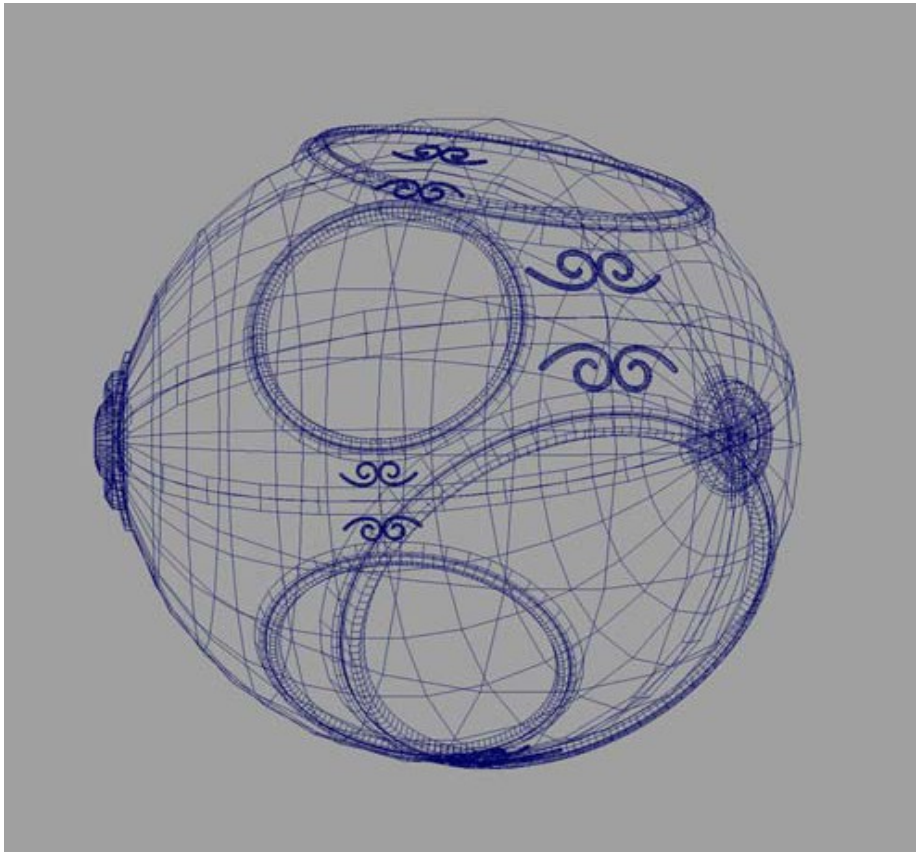
Historical Perspective of Computer Graphics

1. Computer Graphics: 1950-1960

Computer graphics goes back to the earliest days of computing (simple graphics display).

2. Computer Graphics: 1960-1970

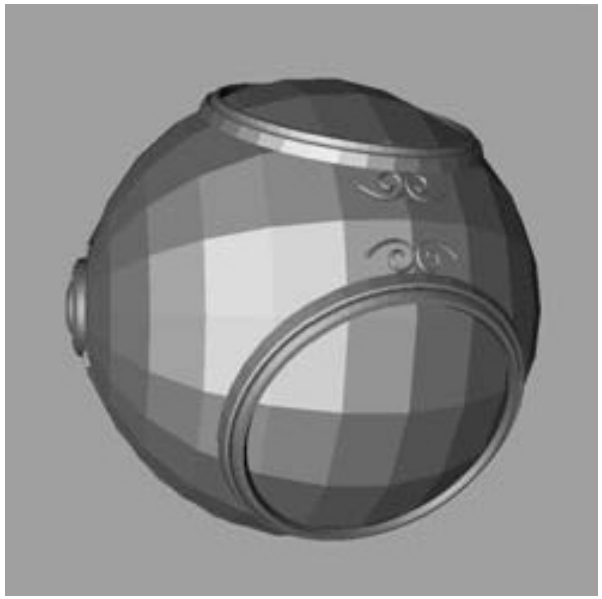
Wireframe graphics (Draw only lines).



3. Computer Graphics: 1970-1980

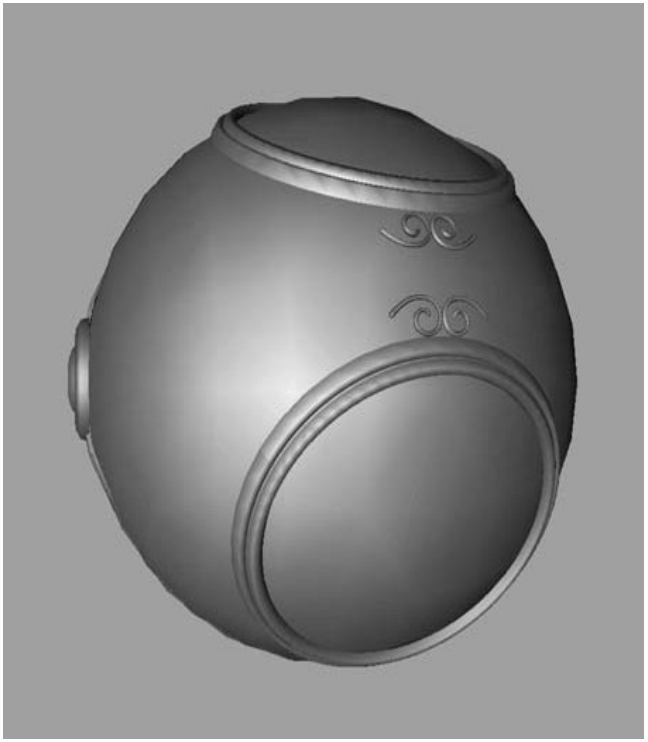
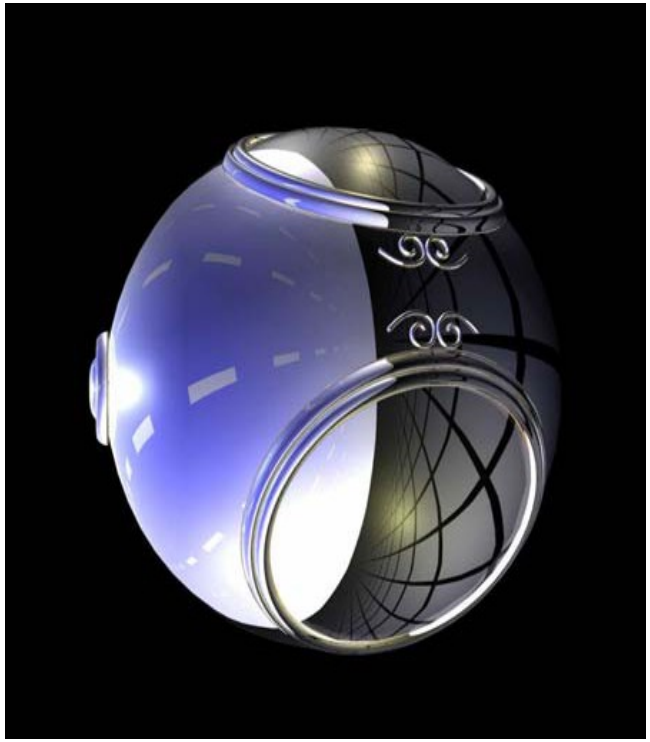
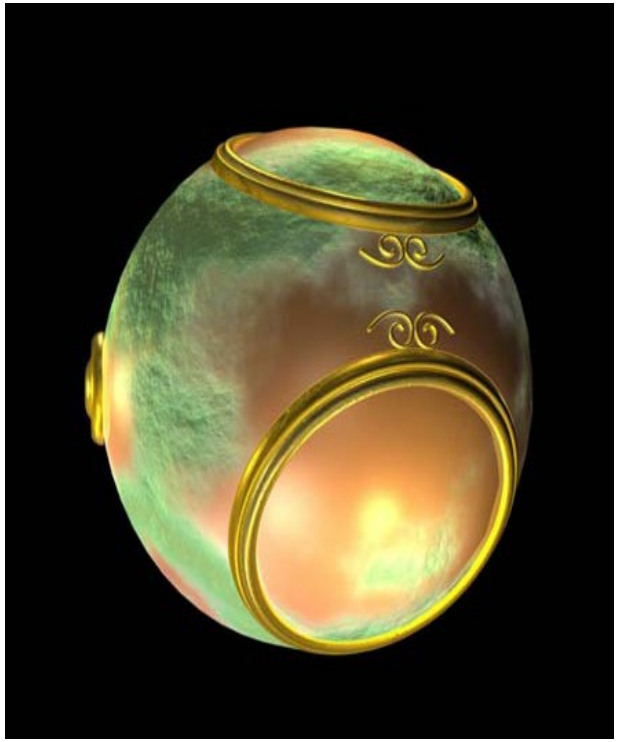
Raster Graphics: Image produced as an array (the raster) of picture elements (pixels) in the frame buffer.

Allows to go from lines and wire frame images to filled polygons



4. Computer Graphics: 1980-1990

Realism comes to computer graphics

Computer Graphics: 1980-1990

Realism comes to computer graphics:

- **Industry-based standards**
- **Networked graphics: X Window System**
- **Human-Computer Interface (HCI)**

5. Computer Graphics: 1990-2000

OpenGL API

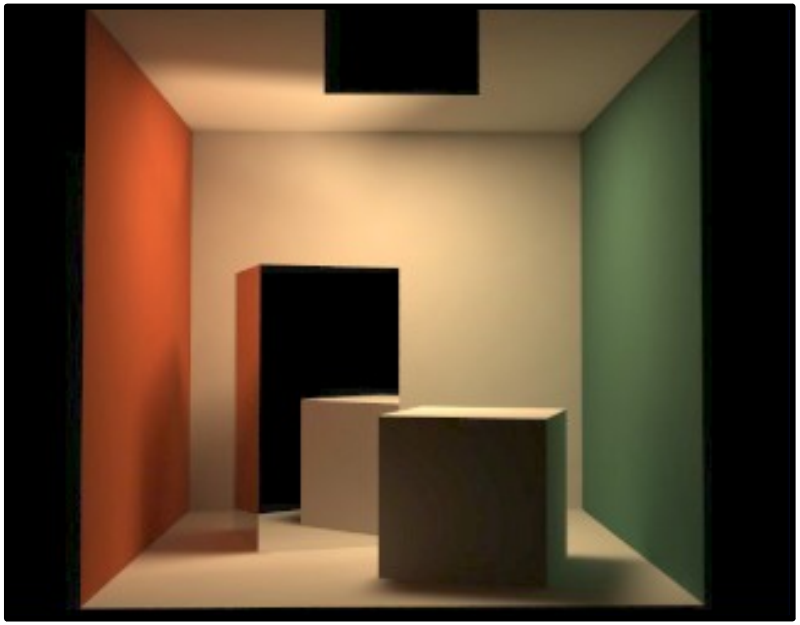
Movies are completely computer generated

New hardware capabilities

6. Computer Graphics: 2000-

Rendering photorealism

Real



Synthetic

Computer Graphics: 2000-

Graphics cards for PCs dominate market

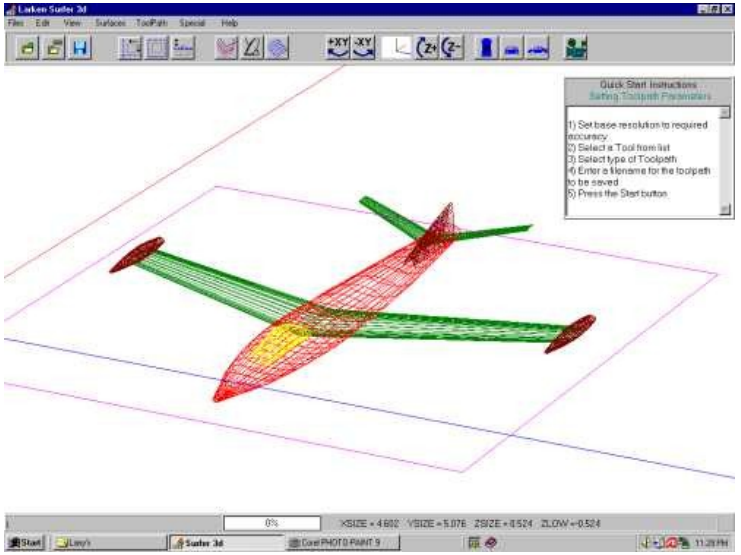
- **GPU (Graphics processing unit)**
- **Computer graphics practice in movie industry**
- **Game industry determine direction of market**

Early Applications of Computer Graphics

- Data Visualization
  - Charts and Graphs

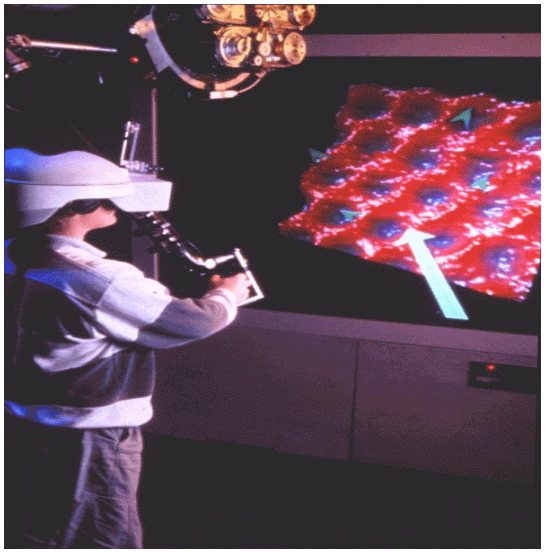
Early Applications of Computer Graphics

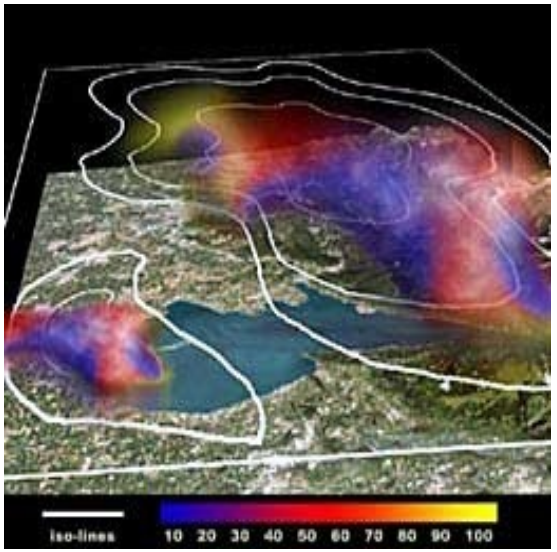
- Computer Aided Design (CAD)



Computer Graphics Applications

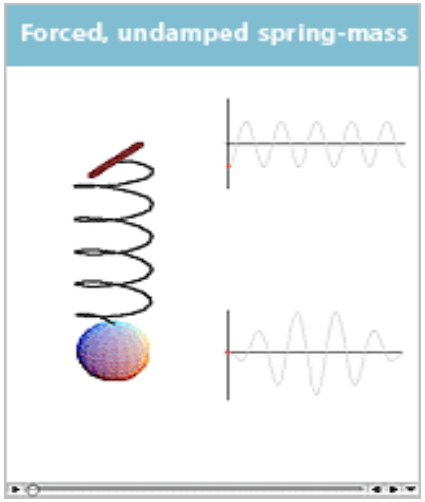
- **Virtual Reality**
  - VR: User interacts and views with a 3D world using “more natural” means
- **Data Visualization**
  - Scientific, Engineering, Medical data
  - Visualizing millions to billions of data points
  - See trends



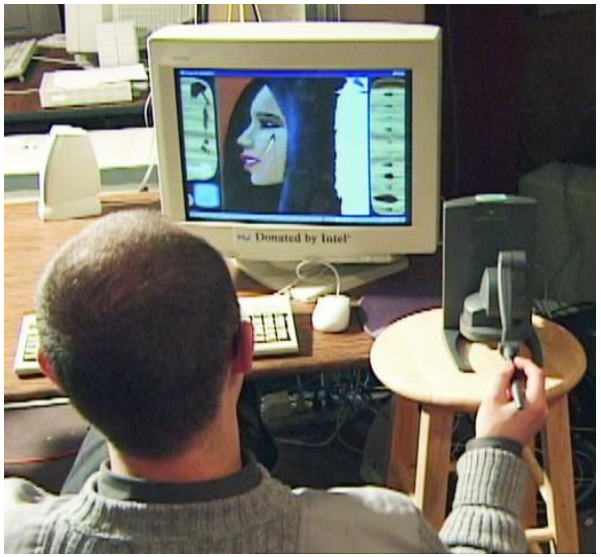


Computer Graphics Applications

- **Education and Training**
  - Models of physical, financial, social systems
  - Comprehension of complex systems



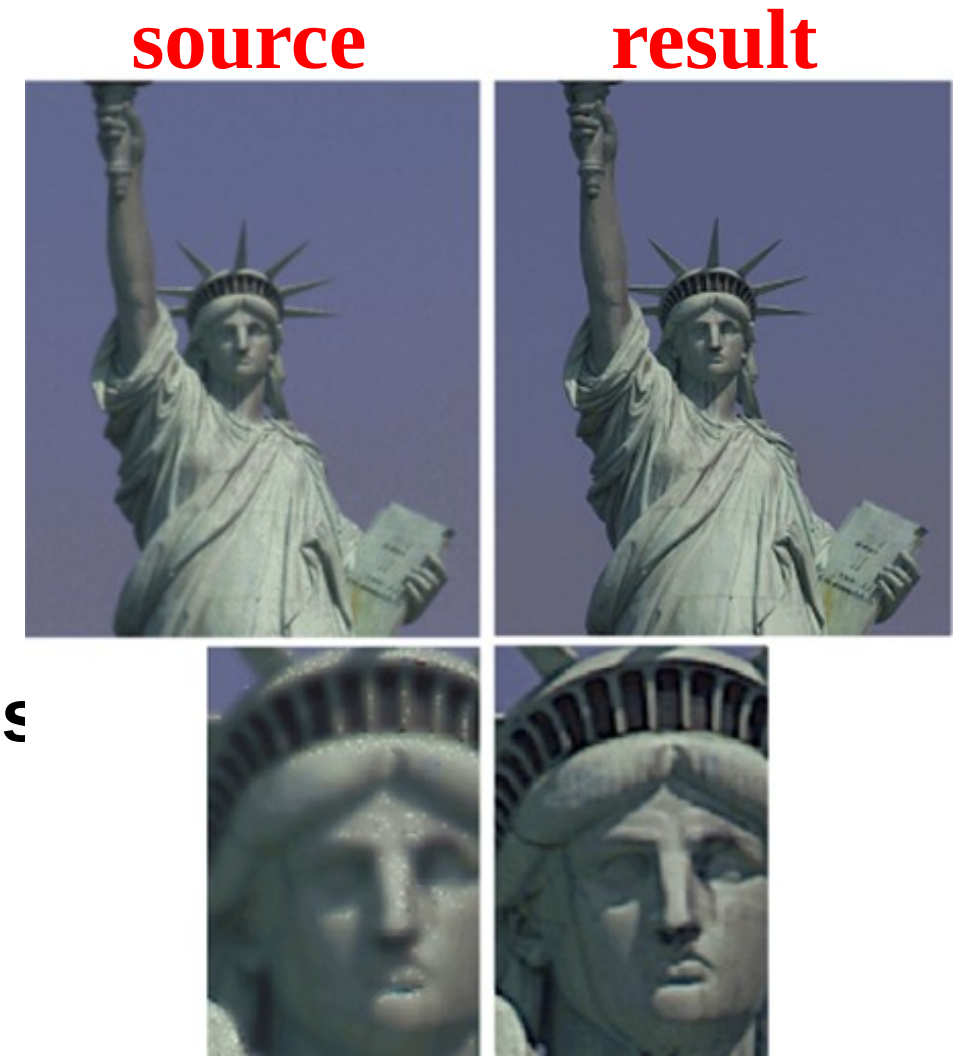
- **Computer Art**
  - Fine and commercial art
  - Performance Art



- **Games/Movies**

Computer Graphics Applications

- **Image Processing**
  - ~Inverse of Graphics
  - Start with a picture
  - Process picture information



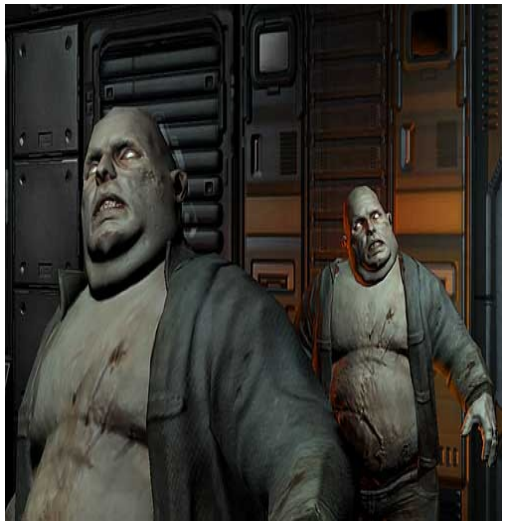
- **Graphical User Interfaces (GUIs)**
  - HCI

What drives computer graphics?

- Movie Industry
  - Leaders in quality and artistry
  - Not slaves to conceptual purity
  - Big budgets and tight schedules
  - How'd they do that?
  - Defines our expectations

What drives computer graphics?

- **Game Industry**
  - **Focus on interactivity**
  - **Cost effective solutions**
  - **Avoiding computation and other tricks**



What drives computer graphics?

- **Medical Imaging and Scientific Visualization**
  - **Tools for teaching and diagnosis**
  - **New data representations and modalities**
  - **Focus on presentation and interpretation of data**
  - **Construction of models from acquired data**
  - **and others**

What drives computer graphics?

- **Computer Aided Design**
  - **Mechanical, Electronic, Architecture,...**
  - **Integration of computing and display resources**
  - **Reduced design cycles == faster systems, sooner**